Regulated Search as Probabilistic Complementary Search

From symbolic search to probabilistic choice, entropy regulation, and self-bootstrap

Core idea

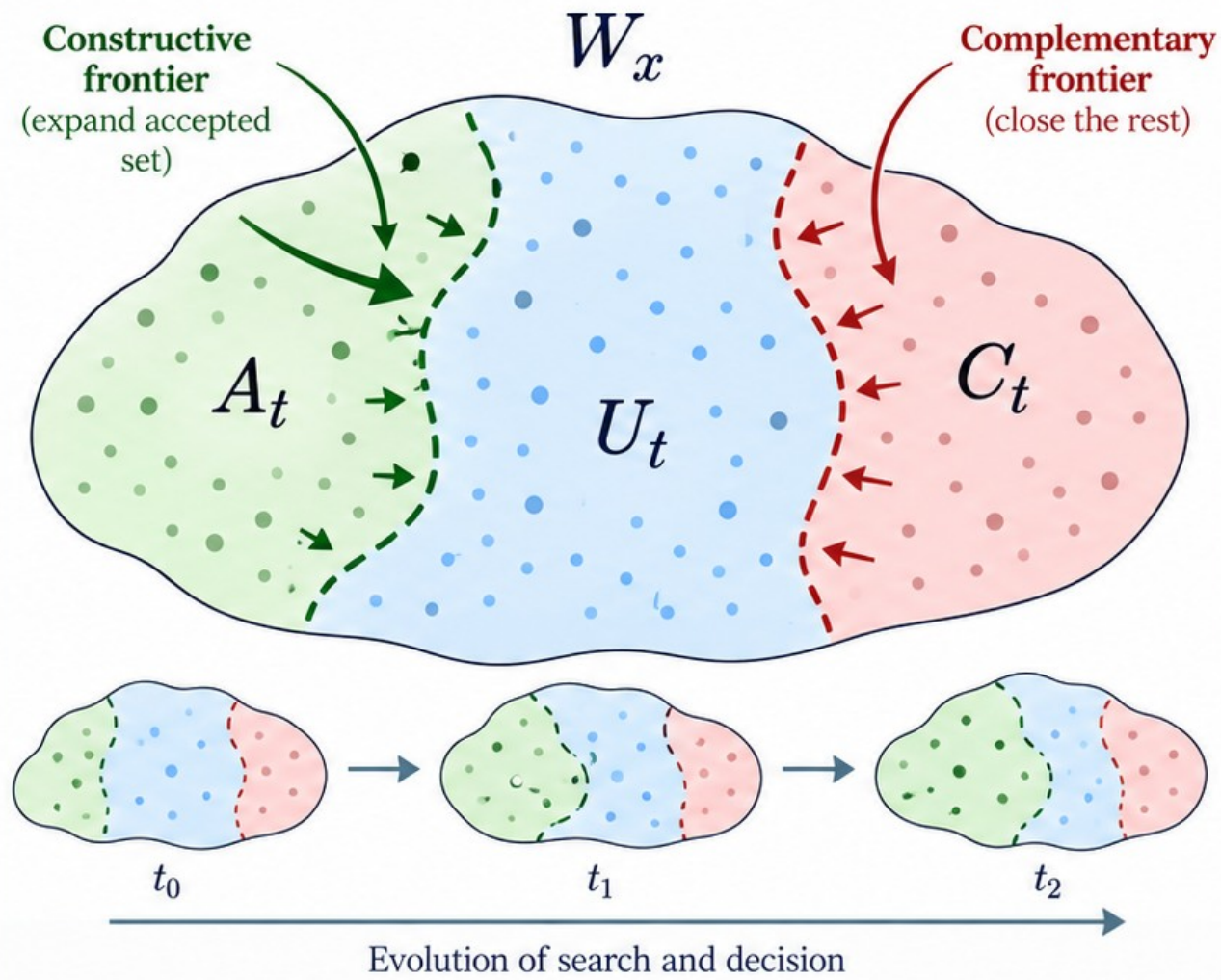
- Search is *not only* generation of candidates.
- It also builds *reasons* for *not* searching the rest.
- Decision is the *progressive resolution* of an unresolved space.
- In the probabilistic version, the unresolved complement carries *probability mass*.

Roadmap

1. Constructive and complementary search
2. Decision cover and branching surplus
3. Probabilistic choice and residual mass
4. Entropy / effective branching
5. MinMax as a limiting case
6. Learning where further search is needed

$$W_x = \underbrace{A_t}_{\text{accepted}} \cup \underbrace{U_t}_{\text{unresolved}} \cup \underbrace{C_t}_{\text{closed}}$$

Search space and complementary frontiers



Constructive and Complementary Search

Two synchronized frontiers over the same unresolved space

Concepts and formulas

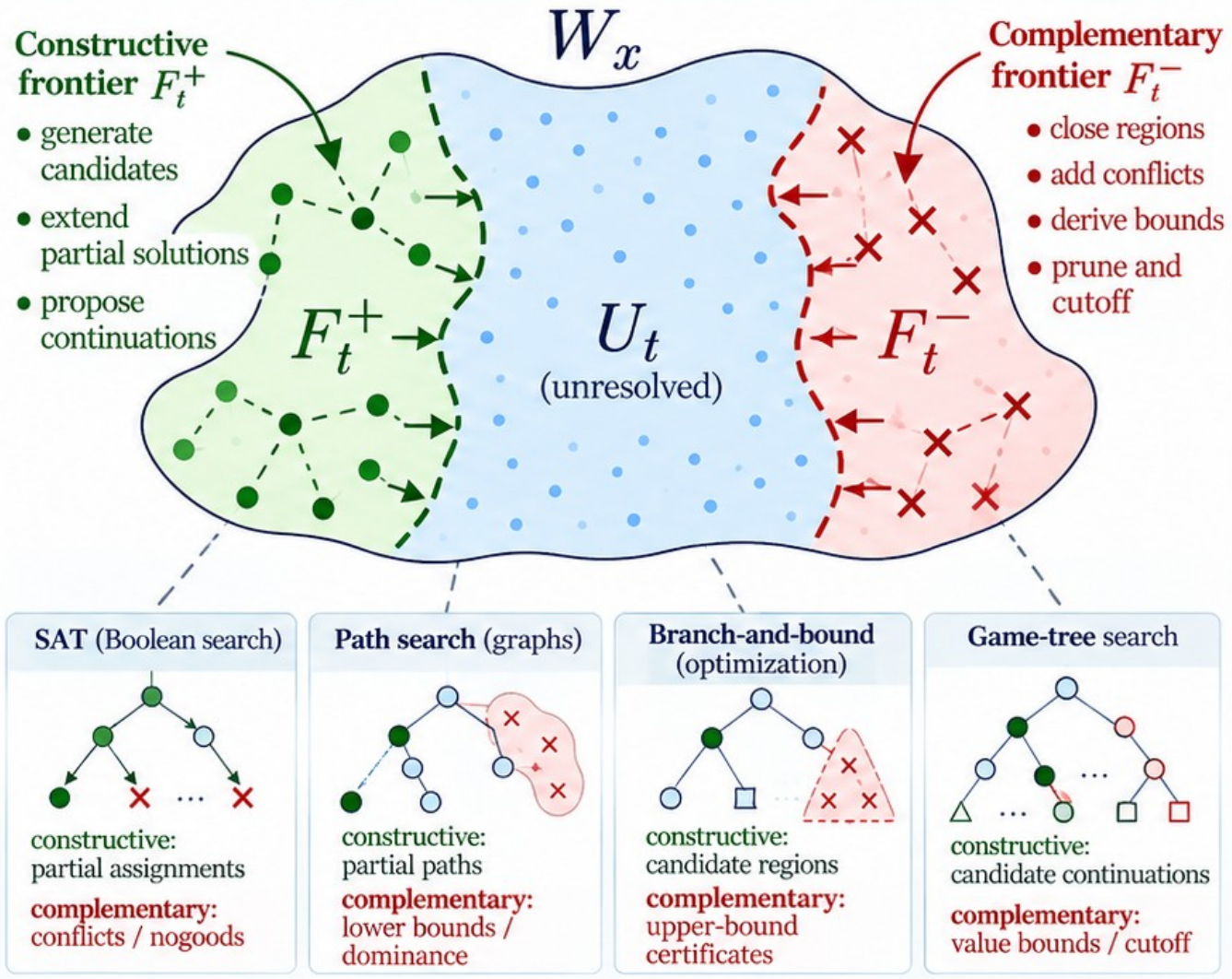
- A decision problem has witness space W_x and verification relation $V(x, w)$ in $\{0,1\}$.
- Constructive search seeks positive evidence: find w such that $V(x, w) = 1$.
- Complementary search builds reasons to close regions that need no further search.
- The two frontiers need not carry the same kind of objects.

$$Sol(x) = \{\, w \in W_x : V(x, w) = 1 \}$$

$$F_t^+ \quad \text{and} \quad F_t^-$$

Constructive object	Complementary object
partial assignment	conflict / nogood
partial path	lower bound / dominance
candidate region	upper-bound certificate
candidate continuation	value bound / cutoff

Search space with two synchronized frontiers



Decision Cover and Branching Surplus

Search as progressive resolution of the unresolved set

Concepts and formulas

- At time t, the witness space is partitioned into accepted, unresolved, and closed regions.
- Soundness requires accepted states to be correct and closed states to contain no solution.
- Regulation balances expansion and closure.

$$W_x = \underset{\text{accepted}}{A_t} \cup \underset{\text{unresolved}}{U_t} \cup \underset{\text{closed}}{C_t}$$

$$A_t \subseteq Sol(x), \quad C_t \cap Sol(x) = \emptyset$$

$$\beta_t = e_t - g_t$$

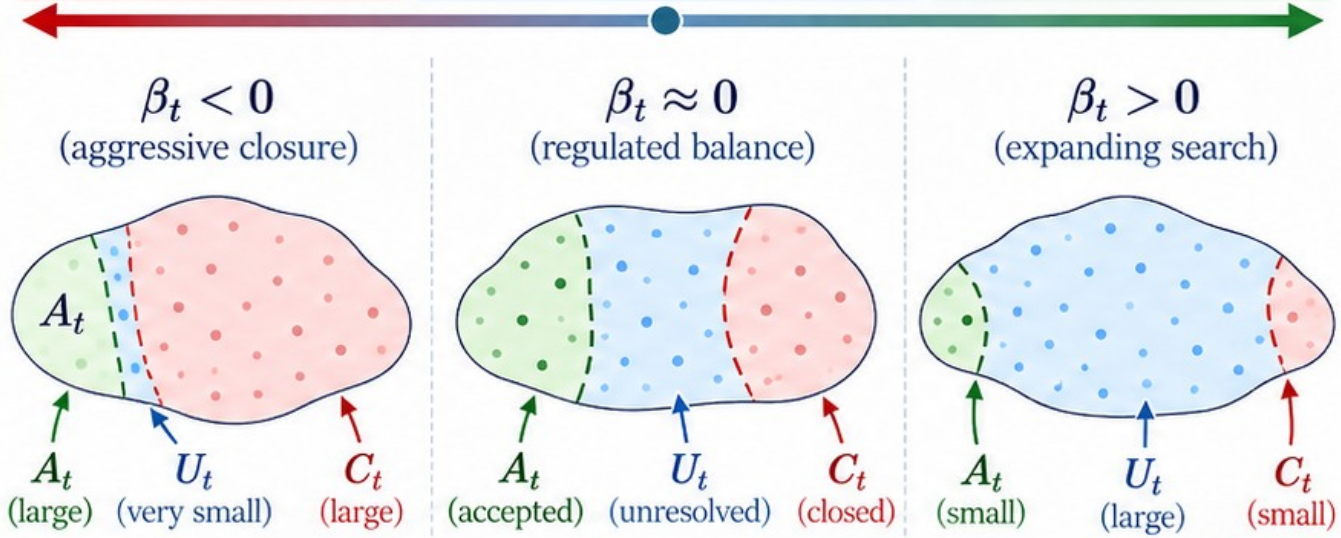
e_t = constructive expansion gain
g_t = complementary closure gain

Implications of the branching surplus

1. $\beta_t > 0$: unresolved space accumulates
2. $\beta_t < 0$: aggressive collapse
3. $\beta_t \approx 0$: regulated metastability

Search regimes and branching surplus

Greedy collapse (very small unresolved) — **Regulated metastability** (balanced search) — **Exhaustive dispersion** (exploding unresolved)



Expansion gain e_t
adds candidates, grows the unresolved set



Closure gain g_t
prunes regions, closes the search space

$$\beta_t = e_t - g_t \approx 0$$

Regulation balances constructive expansion and complementary closure.

The Square-Root Feasible Band

A structural scale for controllable unresolved width

Concepts and formulas

- Let $w(n)$ denote effective unresolved width at problem scale n.
- If maintaining active alternatives costs pairwise interaction, the *relational* cost is $O(w(n)^2)$.
- The scale $w(n) = O(\sqrt{n})$ keeps pairwise maintenance near linear in n.
- In the probabilistic version, effective width is entropy-based.

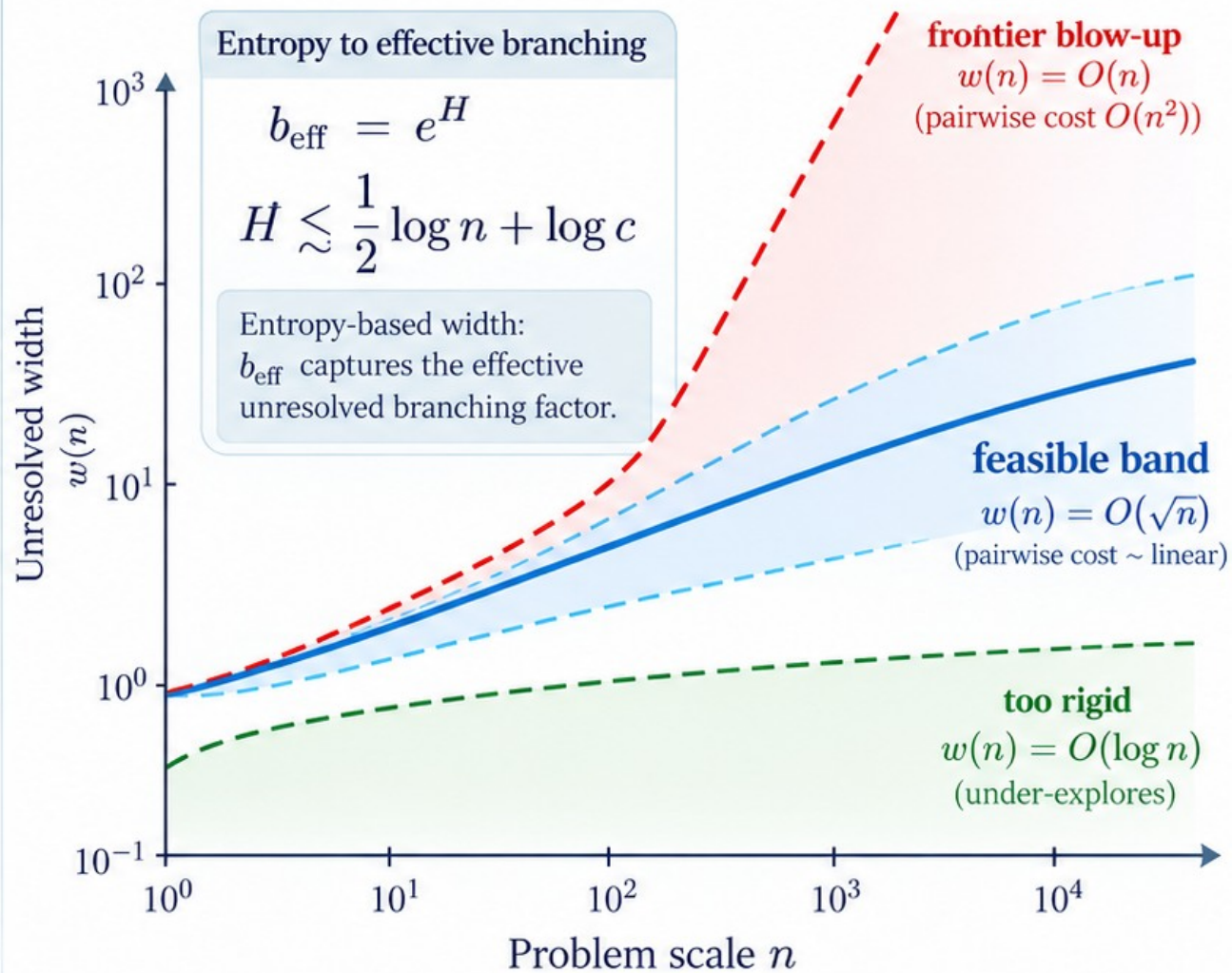
Key formulas

$$
\begin{aligned}
& O(w(n)^2) \\
w(n) &= O(\sqrt{n}) \\
b_{\text{eff}} &= e^{H} \\
H &\lesssim \frac{1}{2}\log n + \log c
\end{aligned}
$$

Caution

The square-root band is a control diagram, not a universal theorem.

Unresolved width vs. problem scale



Probabilistic Choice Search

Decision as a distribution, not only as a best branch

Concepts and formulas

- Replace deterministic choice by a policy $\pi(a \mid s)$.
- The regulator controls concentration of probability mass.
- Entropy and effective branching measure the *true active width*.
- Small temperature approaches deterministic choice.

Action sampled from policy

$$a \sim \pi(\cdot \mid s)$$

Boltzmann policy with temperature τ

$$\pi_\tau(a \mid s) = \frac{\exp(\hat{Q}(s,a)/\tau)}{\sum_b \exp(\hat{Q}(s,b)/\tau)}$$

Policy entropy (measures uncertainty)

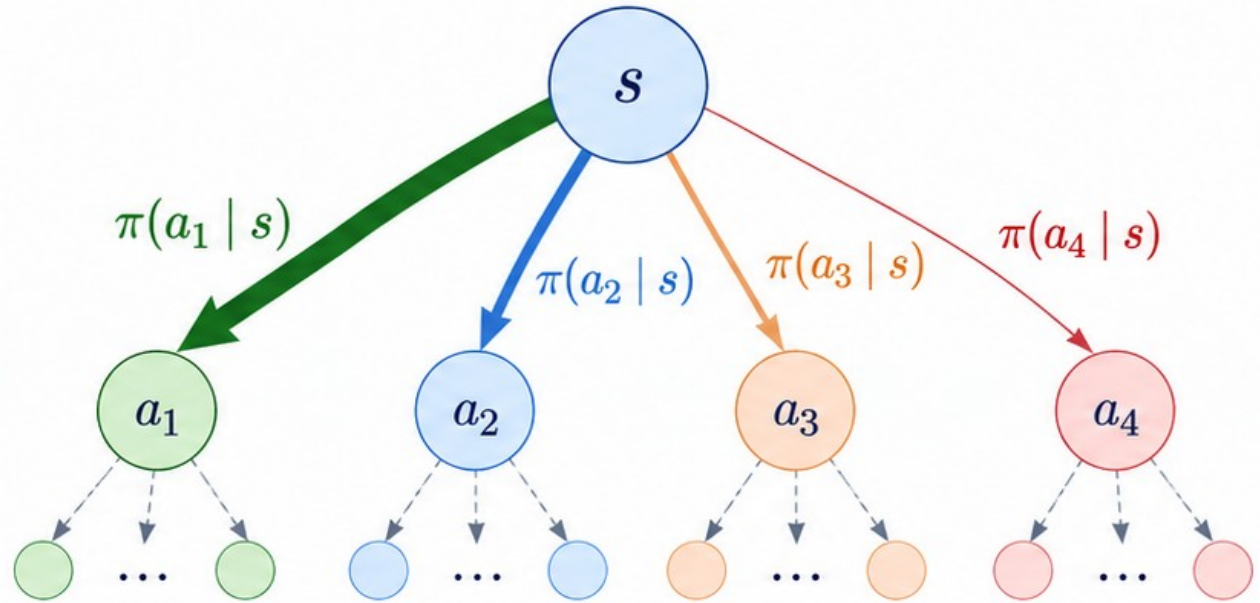
$$H(s) = -\sum_a \pi(a \mid s) \log \pi(a \mid s)$$

Effective branching (number of active choices)

$$b_{eff}(s) = e^{H(s)}$$

τ small $\rightarrow$ concentrated policy (near-deterministic)
τ large $\rightarrow$ diffuse policy (explores many actions)

Decision tree with probabilistic choice



Temperature τ (controls concentration)

diffuse policy (explore) — large τ — small τ — concentrated policy (exploit)

Entropy $H(s)$ (measures active width)

high entropy (many choices active) — low entropy (few choices active)

Residual Probability and Stopping

The unsearched complement carries measurable decision risk

Concepts and formulas

- Let E_t be the searched set of actions or continuations.
- The unresolved *complement* U_t carries *residual probability mass.*
- Truncation becomes quantitative: we can bound the value still hidden in the unresolved complement.
- Search may stop when the unresolved contribution is small enough.

Key formulas

$$\rho_t = \sum_{a \in U_t} \pi(a \mid s) \qquad \text{(residual probability mass)}$$

$$V(s) \in [V_E + L_t^R,\ V_E + U_t^R] \qquad \text{(value interval)}$$

$$\Delta_t^R = U_t^R - L_t^R \qquad \text{(interval width)}$$

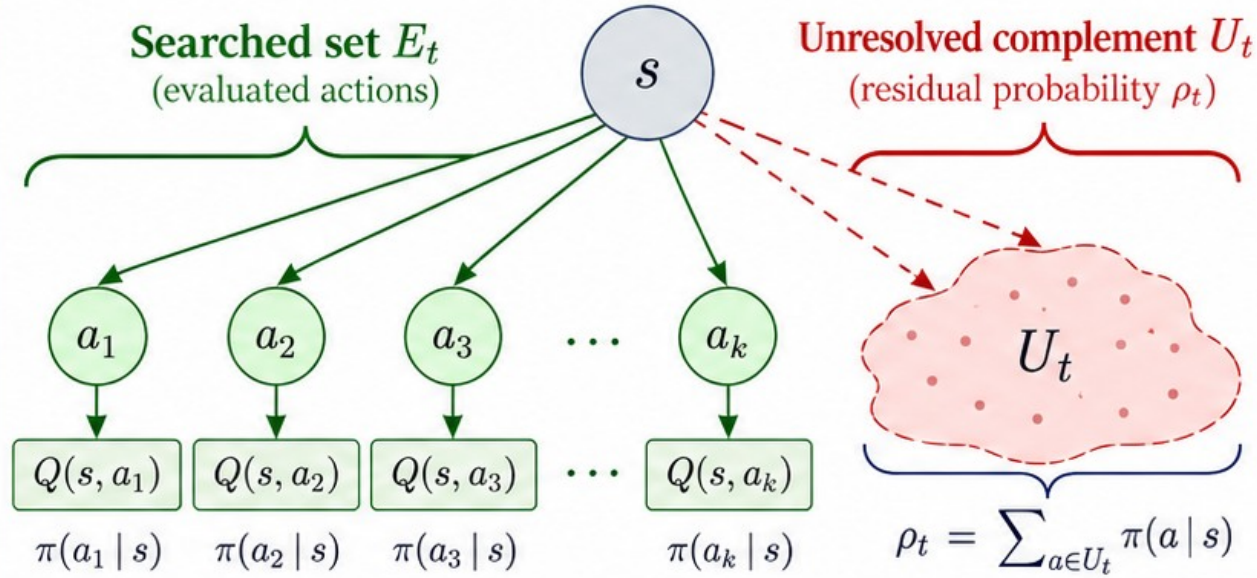
$$\Delta_t^R \le \rho_t (U - L) \qquad \text{(bound from residual mass)}$$

$$\rho_t (U - L) < \delta \implies \text{stop} \qquad \text{(stopping condition)}$$

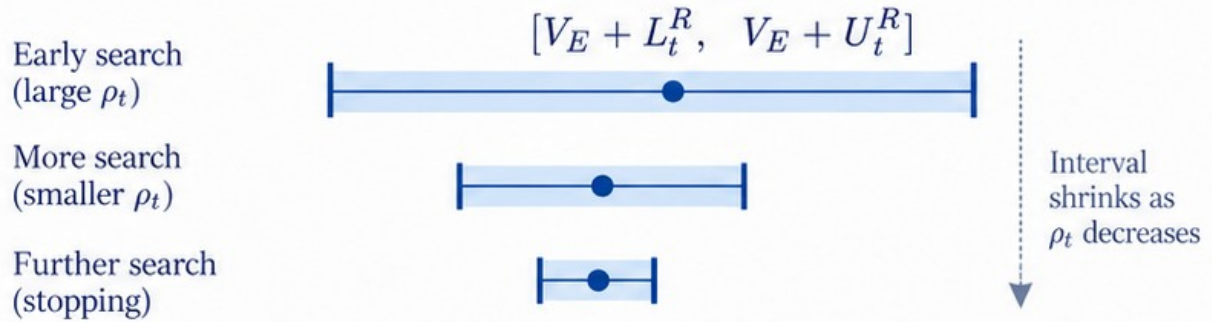
Plain-language criterion

Search stops when unresolved probability mass cannot change the decision by more than delta.

Searched set and residual complement



Value interval shrinks as search progresses



MinMax and Alpha-Beta as a Limiting Case

Deterministic adversarial search inside the regulated framework

Concepts and formulas

- MinMax assumes an adversarial opponent choosing the worst continuation.
- Probabilistic choice approaches deterministic choice as temperature goes to zero.
- Alpha-beta pruning is exact complementary closure.
- Better move ordering improves closure earlier.

Key formulas

$$V(s) = \max_a \min_b V(\mathcal{T}(\mathcal{T}(s,a),b))$$

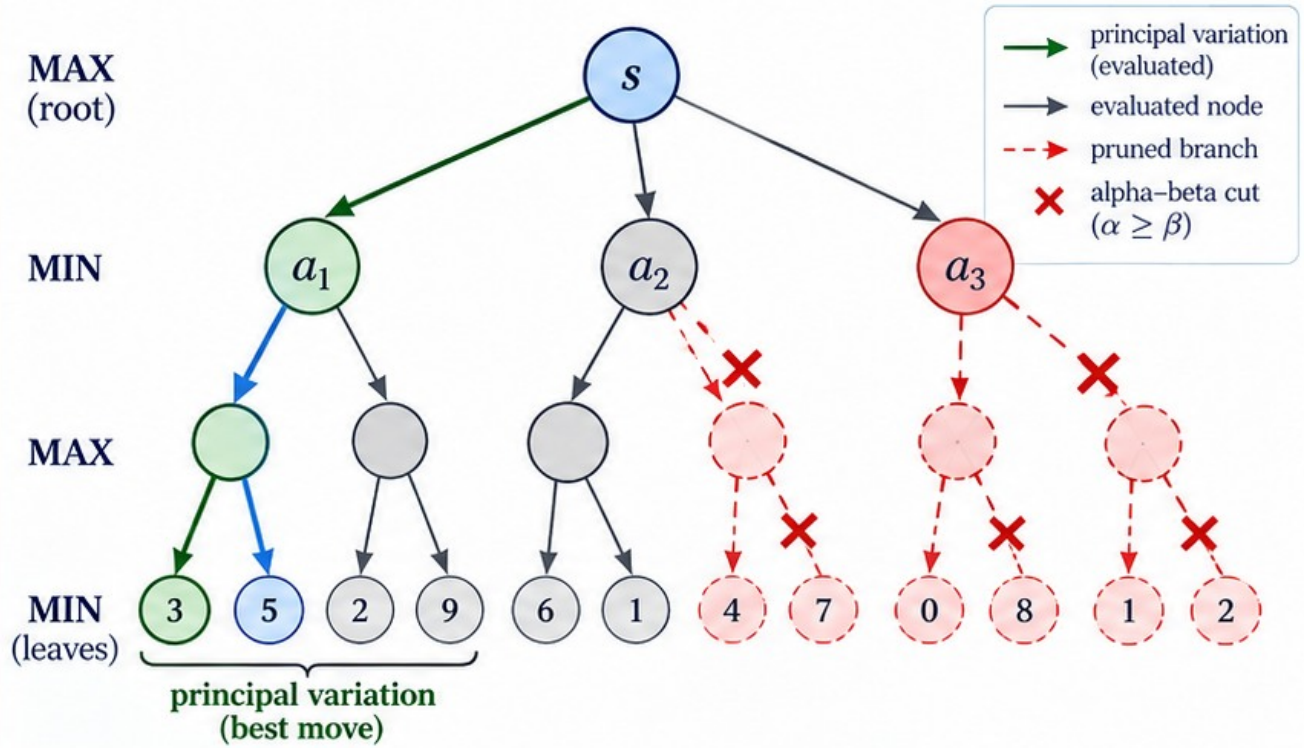
$$\tau \to 0 \quad \Rightarrow \quad \pi(a^* \mid s) \to 1$$

$$\alpha \geq \beta \quad \Rightarrow \quad \text{close branch}$$

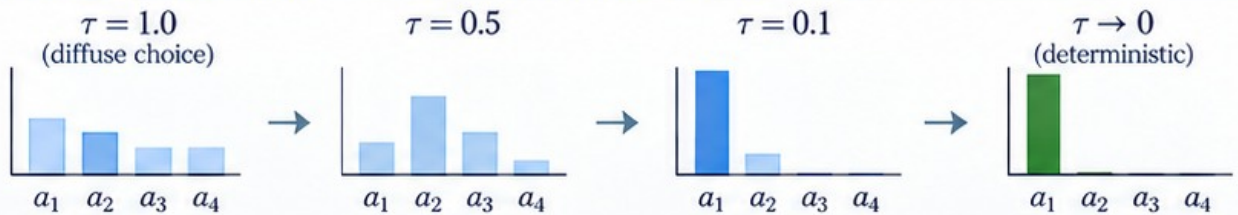
From ordering to deeper search

1 better ordering → 2 stronger bounds → 3 more closure → 4 deeper search

Adversarial game tree and alpha-beta pruning



Probabilistic choice → deterministic choice as $\tau \to 0$



Learning Where Search Is Needed

Evaluation predicts value; regulation predicts need for further search

Concepts and formulas

- A *value predictor* estimates the worth of a state.
- A *residual predictor* estimates how dangerous it is to stop searching.
- Regulation should allocate computation where uncertainty still matters.
- Keep evaluation and closure conceptually distinct.

$$\widehat{V}_\theta(s) \approx V(s)$$

$$r(s) = V_{search}(s) - \widehat{V}_\theta(s)$$

$$\widehat{R}_\phi(s) \approx |r(s)|$$

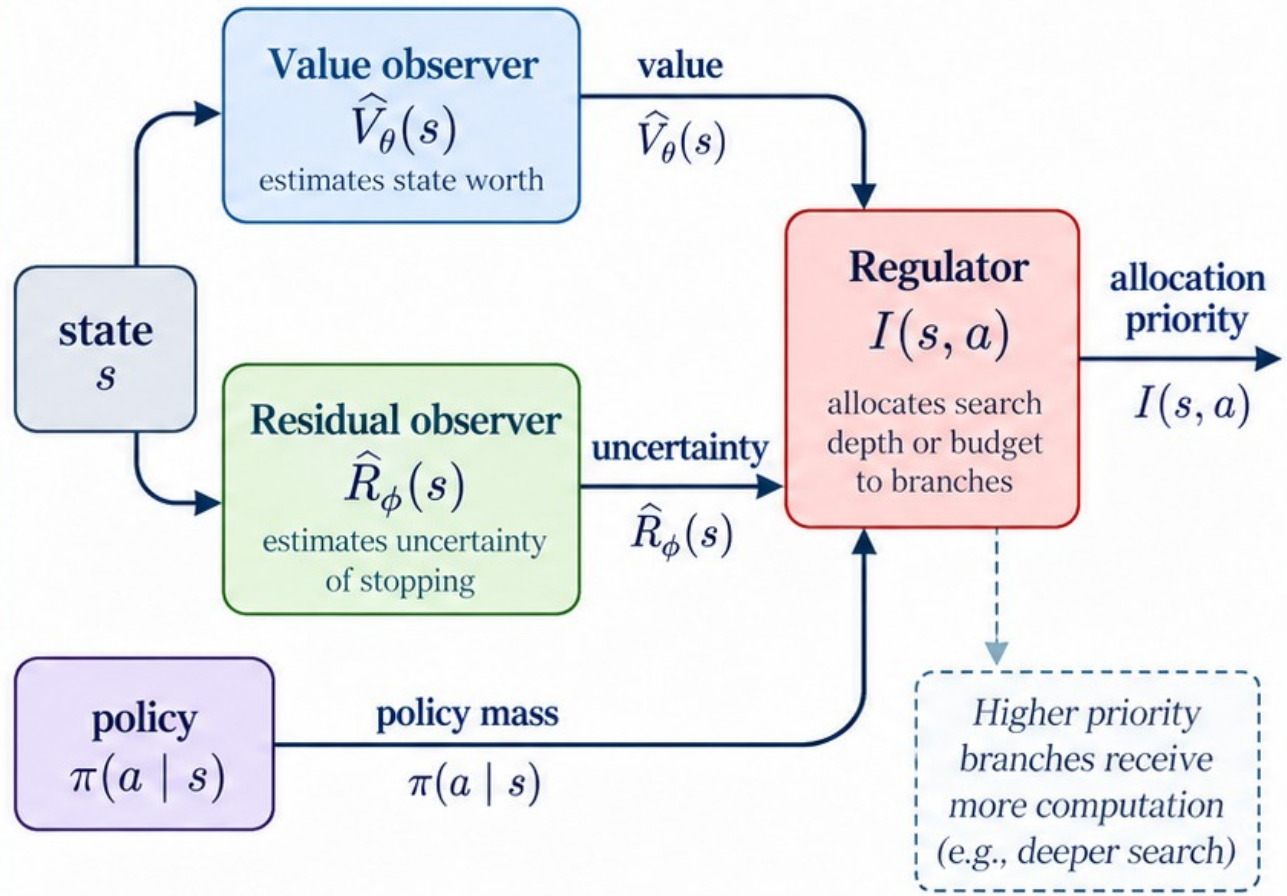
$$I(s,a) = \pi(a \mid s)\, \widehat{R}_\phi(s,a)$$

Key distinction

evaluation: what is this state worth?

regulation: where should more search be spent?

Learning architecture for guided search



Regulator and Self-Bootstrapping Architecture

A closed loop that improves both search and prediction

Concepts and formulas

- The regulator maps observations of the search process into control decisions.
- Typical controls: candidate width, local depth, quiescence, stopping threshold.
- Deeper regulated search can generate targets for cheaper predictors.
- The same architecture acts as student and teacher under different budgets.

$$\mathcal{O}_t \mapsto \mathcal{U}_t$$

$$\mathcal{O}_t = (|F_t|,\ \beta_t,\ H_t,\ \rho_t,\ U_t)$$

frontier size; branching factor; search horizon; residual mass; unresolved set size

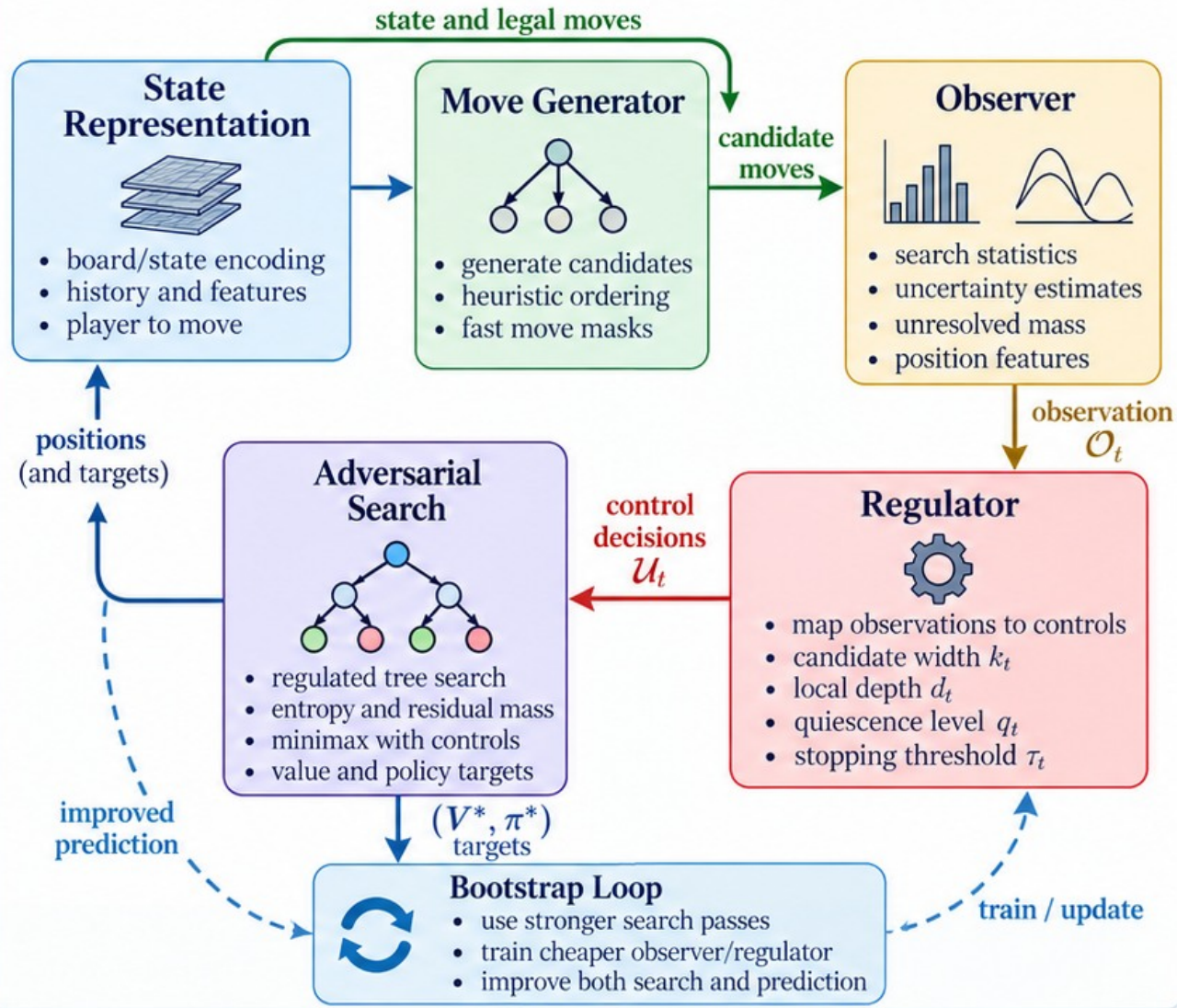
$$\mathcal{U}_t = (k_t,\ d_t,\ q_t,\ \tau_t)$$

candidate width; local depth; quiescence level; stopping threshold

Bootstrap chain

positions → deeper regulated search → (V^*, π^*) → update observer and regulator

Architecture and learning loop



Proof of Concept and Main Takeaways

A small experiment and the core message of the framework

Experiment and takeaways

Experiment

Domain: 4x4 Connect Four
4500 random nonterminal positions
160 held-out early/mid-game test positions
Node budget: 80

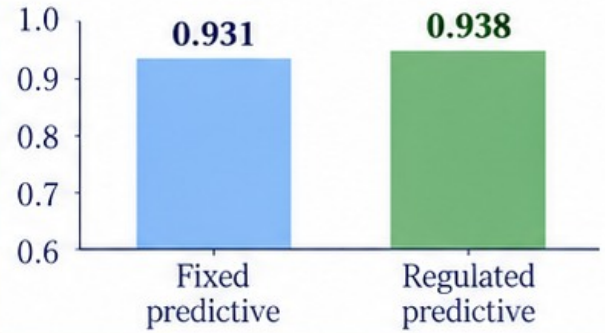
Agent	Optimal move	Mean regret	Mean nodes
Fixed predictive	0.931	0.069	50.7
Regulated predictive	**0.938**	**0.062**	**40.2**

- Regulated prediction achieved slightly better decisions with fewer nodes.
- The key gain came from learning *need for search, not* only value.
- Residual probability and entropy provide measurable control targets.

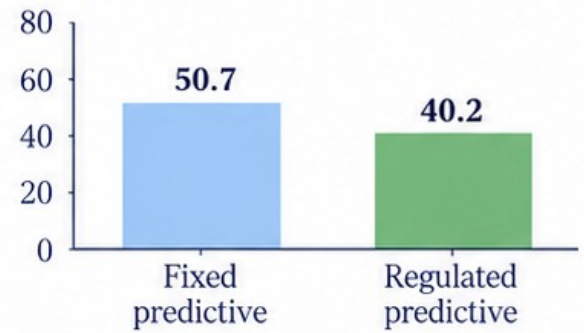
Search becomes powerful when it learns both how to generate promising possibilities and how to regulate the unresolved rest.

Comparison and overall picture

Optimal-move rate (higher is better)



Mean nodes (lower is better)



From prediction to regulated complementary search

